CLASS 10 - SCIENCE

Faculty Evaluation Reference Pack

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Exam summary

Subject	Science
Class	10
Maximum marks	25
Questions	7
Purpose	Model answers, key concepts, acceptable alternatives, reference solutions and marking notes for AI-assisted evaluation.

Q1 - 3 marks

Question: Explain why a metal spoon feels colder than a wooden spoon when both are kept in the same room for a long time.

Model answer: Both spoons are at approximately the same room temperature. Metal conducts heat away from the hand much faster than wood, so the hand loses heat more quickly and the metal feels colder.

Key concepts: thermal conductivity; heat transfer from hand; same ambient temperature

Acceptable alternatives: Any explanation stating that metal is a better conductor than wood and therefore removes heat from the hand faster.

Reference solution: 1) Both objects reach room temperature. 2) On touching them, heat flows from the warmer hand to the spoon. 3) Metal has much higher thermal conductivity than wood. 4) Faster heat loss from the hand creates the sensation of greater coldness.

Marking notes: Award 1 mark for noting same/similar room temperature, 1 mark for metal being a better conductor, and 1 mark for linking faster heat transfer from the hand to the sensation of cold.

Q2 - 4 marks

Question: A 12 V battery is connected across a 6 ohm resistor. Calculate (a) the current through the resistor and (b) the electrical power consumed.

Model answer: Using Ohm's law, I = V/R = 12/6 = 2 A. Power P = VI = 12 x 2 = 24 W.

Key concepts: Ohm's law; I = V/R; electrical power; P = VI or V^2/R

Acceptable alternatives: Power may also be calculated as P = I^2R = 2^2 x 6 = 24 W or P = V^2/R = 144/6 = 24 W.

Reference solution: Current: I = 12/6 = 2 A. Power: P = 12 x 2 = 24 W.

Marking notes: 2 marks for correct current with method; 2 marks for correct power with method. Give partial credit for correct formula even if arithmetic has a minor error.

Q3 - 3 marks

Question: State two differences between aerobic respiration and anaerobic respiration, and give one example of each.

Model answer: Aerobic respiration requires oxygen and releases more energy; anaerobic respiration occurs without oxygen and releases less energy. Example aerobic: respiration in human cells under normal conditions. Example anaerobic: fermentation by yeast or lactic acid formation in muscles during vigorous exercise.

Key concepts: oxygen requirement; relative energy release; examples

Acceptable alternatives: Any scientifically correct examples of aerobic and anaerobic respiration are acceptable.

Reference solution: Difference 1: oxygen required vs not required. Difference 2: greater vs lower energy yield (or complete vs incomplete breakdown). Examples: human cellular respiration / yeast fermentation.

Marking notes: 1 mark for each valid difference (2 marks total) and 1 mark for valid examples.

Q4 - 4 marks

Question: A student places an iron nail in copper sulphate solution. Describe two observations and write the balanced chemical equation for the reaction.

Model answer: The blue copper sulphate solution gradually turns green because iron sulphate forms, and a reddish-brown deposit of copper appears on the iron nail. Fe + CuSO4 -> FeSO4 + Cu.

Key concepts: displacement reaction; colour change blue to green; copper deposition; balanced equation

Acceptable alternatives: Descriptions such as copper coating/deposit on nail and fading/green colour of solution are acceptable.

Reference solution: Observation 1: blue solution becomes green. Observation 2: reddish-brown copper deposits on iron. Equation: Fe + CuSO4 -> FeSO4 + Cu.

Marking notes: 1 mark per correct observation (2 marks), 2 marks for correct balanced equation including products.

Q5 - 3 marks

Question: Why does the sky appear blue during the day? Explain using the scattering of light.

Model answer: Sunlight contains many colours. Shorter wavelengths such as blue light are scattered more strongly by the small particles and molecules in the atmosphere than longer wavelengths. Therefore more blue light reaches our eyes from different directions and the sky appears blue.

Key concepts: scattering; shorter wavelength; blue scattered more than red; atmosphere

Acceptable alternatives: Rayleigh scattering explanation in equivalent wording is acceptable.

Reference solution: White sunlight enters atmosphere -> shorter wavelengths scatter more strongly -> scattered blue light reaches observer from all directions -> sky appears blue.

Marking notes: Award 1 mark for scattering by atmosphere, 1 mark for stronger scattering of shorter wavelengths, and 1 mark for linking this to the blue appearance.

Q6 - 4 marks

Question: A plant is kept in the dark for 24 hours and then one leaf is partly covered with black paper before placing the plant in sunlight. After a few hours, the leaf is tested with iodine. What result is expected and what does the experiment show?

Model answer: The exposed part of the leaf turns blue-black with iodine, while the covered part remains brown/yellow. The experiment shows that light is necessary for photosynthesis and starch formation.

Key concepts: destarching; iodine starch test; exposed part blue-black; covered part no starch; light necessary for photosynthesis

Acceptable alternatives: Equivalent description of starch forming only in the illuminated region is acceptable.

Reference solution: Destarched plant -> only illuminated region photosynthesizes -> starch forms there -> iodine turns illuminated area blue-black; covered area stays brown/yellow.

Marking notes: 2 marks for correct iodine-test observations and 2 marks for correct conclusion.

Q7 - 4 marks

Question: Explain how a fuse protects an electrical circuit. Why is the fuse wire connected in series rather than in parallel?

Model answer: A fuse contains a wire with a low melting point. If current exceeds the safe value, heating causes the fuse wire to melt and break the circuit, stopping current flow. It is connected in series so that all circuit current passes through it and melting the fuse opens the complete circuit.

Key concepts: excess current; heating effect; low melting point; circuit breaks; series connection

Acceptable alternatives: MCB comparison is acceptable only if the role of the fuse itself is still explained correctly.

Reference solution: High current -> heating -> fuse melts -> circuit opens. Series connection ensures the full current passes through the fuse and that melting interrupts the supply.

Marking notes: 2 marks for protective action, 1 mark for low melting point/heating explanation, 1 mark for why series connection is required.